Assignment 1 - Medium

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1 Instructions

This assignment is based on Lectures 1-6: bond features and risks, pricing and yield measures, duration and convexity, benchmark spreads and term structure, Treasury and corporate debt markets, and residential mortgages/MBS.

The questions are original CFA-style practice questions written from the course material. They are not copied from CFA Institute books or exams.

Questions 1-10 are conceptual, Questions 11-20 are calculations, and Questions 21-25 use the generated spreadsheet files in this folder.

1.1 Question 1 (Conceptual)

A bank portfolio manager says, 'I prefer floaters because they have no interest-rate risk.' Evaluate the statement.

Answer. The statement is too strong. Floaters have lower price sensitivity to small parallel shifts shortly after reset because coupons adjust to current short rates, but they still carry spread risk, liquidity risk, cap/floor risk, reset-lag risk, and credit risk. A floater can lose value if the issuer's required margin widens even when SOFR is unchanged.

1.2 Question 2 (Conceptual)

Why can a lower-coupon bond have higher duration than a higher-coupon bond with the same maturity?

Answer. Lower coupons push more of the bond's value into the final principal payment. Because more present value is received later, the weighted average timing of cash flows is later and the price is more sensitive to discount-rate changes. Higher-coupon bonds return more value earlier, which shortens duration.

1.3 Question 3 (Conceptual)

Explain why pull to par happens when yields are unchanged.

Answer. As maturity approaches, fewer cash flows remain and the principal repayment is less heavily discounted. A discount bond rises toward par because the below-market coupon matters for fewer periods. A premium bond falls toward par because the above-market coupon advantage is being used up.

1.4 Question 4 (Conceptual)

Why does a callable bond tend to have lower positive convexity than an otherwise identical option-free bond?

Answer. When rates fall, the option-free bond's price can rise substantially. The callable bond's price is capped by the increasing probability of being called, so the upside is limited. The issuer's embedded option therefore subtracts value from the investor and can create negative convexity in the rate region where the call becomes likely.

1.5 Question 5 (Conceptual)

Distinguish default risk, downgrade risk, and credit spread risk.

Answer. Default risk is failure to make promised payments. Downgrade risk is the risk that a rating agency lowers the issuer's rating, possibly causing forced selling and higher funding costs. Credit spread risk is the broader market repricing of compensation for credit and liquidity risk; it can occur without default or an actual downgrade.

1.6 Question 6 (Conceptual)

Why does the on-the-run Treasury curve not always equal the theoretical risk-free discount curve?

Answer. On-the-run Treasuries are the newest and most liquid benchmark issues, so they often trade rich and yield slightly less than comparable seasoned securities. That liquidity premium can contaminate the curve if the goal is pure discounting. A theoretical spot curve tries to infer zero-coupon discount rates across maturities from many prices while controlling for such distortions.

1.7 Question 7 (Conceptual)

Explain the economic meaning of a forward rate derived from spot rates.

Answer. A forward rate is the break-even reinvestment rate implied by today's spot rates. If the forward rate from year 1 to year 2 is 5.6%, an investor is indifferent, before risk premiums, between investing for two years directly and investing for one year then reinvesting for the second year at that forward rate. It is not automatically a forecast; it can include term and liquidity premiums.

1.8 Question 8 (Conceptual)

Why do market-cap-weighted bond indexes have a structural weakness?

Answer. They give the largest weights to the largest borrowers, not necessarily to the safest or most attractive issuers. A heavily indebted issuer can become a larger benchmark constituent precisely because it has issued more debt. This creates an incentive for benchmark-aware managers to hold more exposure to large debtors unless they actively deviate.

1.9 Question 9 (Conceptual)

Why is average life important for a mortgage pool but insufficient by itself?

Answer. Average life summarizes the weighted timing of principal repayments, which is useful for comparing mortgage pools with prepayments. But two pools can share the same average life while having different month-by-month cash-flow distributions, prepayment convexity, and extension risk. Investors need the full cash-flow path, not only one summary statistic.

1.10 Question 10 (Conceptual)

What makes a leveraged loan different from a high-yield bond?

Answer. Leveraged loans usually have floating rates, are often senior secured, and sit higher in the capital structure. High-yield bonds usually have fixed coupons, longer maturities, and may be unsecured. The loan has less rate duration but still has credit, liquidity, covenant, and refinancing risk.

1.11 Question 11 (Calculation)

A 6-year bond has \$1,000 face value, 6% annual coupon paid semiannually, and a 5.20% yield. Compute the price.

Answer. Semiannual coupon = \$30, semiannual yield = 2.60%, and there are 12 periods. Price = sum of discounted coupons plus principal = \$1,040.78. The bond trades above par because its coupon rate exceeds its yield.

1.12 Question 12 (Calculation)

Compute Macaulay and modified duration for a 4-year annual-pay 5% coupon bond yielding 6%.

Answer. Using present-value weights for the four cash flows, Macaulay duration is 3.718 years. Modified duration = Macaulay duration/(1 + y) = 3.507. A 1 bp rise in yield lowers price by roughly 0.0351%.

1.13 Question 13 (Calculation)

For the bond in Question 12, use modified duration and convexity to approximate the percentage price change for a 75 bp yield increase.

Answer. Convexity is approximately 16.133. The approximation is -3.507 x 0.0075 + 0.5 x 16.133 x 0.0075^2 = -2.585%. Dollar change is about -24.96.

1.14 Question 14 (Calculation)

A 5-year annual-pay bond has a 4% coupon and 5.50% yield. Compute price and identify discount or premium.

Answer. Price = \$935.95. It is a discount bond because the coupon rate is lower than the required yield.

1.15 Question 15 (Calculation)

Price a 3-year annual-pay 4% Treasury with \$1,000 face value using spot rates of 3.50%, 4.00%, and 4.50%.

Answer. Price = \$40/1.035 + \$40/1.04^2 + \$1,040/1.045^3 = \$986.98. Each cash flow is discounted at the spot rate matching its date.

1.16 Question 16 (Calculation)

A 1-year zero priced at 96.1538 implies z1 = 4.00%. A 2-year 5% annual coupon Treasury with \$100 face value is priced at 99.4380. Bootstrap z2.

Answer. Solve 99.4380 = 5/1.04 + 105/(1 + z2)^2. This gives z2 = 5.34%.

1.17 Question 17 (Calculation)

The 1-year spot rate is 4.00% and the 2-year spot rate is 4.80%. Compute the one-year forward rate from year 1 to year 2.

Answer. 1 + f = (1.048)^2/1.04, so f = 5.61%. This is the break-even second-year reinvestment rate.

1.18 Question 18 (Calculation)

A 182-day T-bill has $10,000 face value and a 5.20% bank discount yield. Compute price and bond-equivalent yield.

Answer. Price = $10,000 x [1 - 0.052 x 182/360] = $9,737.11. BEY = (Face - Price)/Price x 365/182 = 5.41%.

1.19 Question 19 (Calculation)

A corporate bond is quoted clean at 101.25% of par with $1,000 face value. Accrued interest is $17.50. Compute full price.

Answer. Clean dollar price is 101.25% x $1,000 = $1,012.50. Full price = $1,012.50 + $17.50 = $1,030.00.

1.20 Question 20 (Calculation)

A mortgage pool has beginning balance $750,000, scheduled principal $1,200, and CPR of 9%. Compute SMM and expected prepayment.

Answer. SMM = 1 - (1 - 0.09)^(1/12) = 0.7828%. Prepayment = SMM x ($750,000 - $1,200) = $5,861.92.

1.21 Question 21 (Spreadsheet)

Open `medium_portfolio_duration.csv`. Compute market-value weights and portfolio duration.

Answer. Weights are market value divided by total market value. Total market value is $4,850,000. Portfolio duration is the weighted average: approximately 4.99 years.

1.22 Question 22 (Spreadsheet)

Open medium_bootstrap.csv. Bootstrap the 1-year, 2-year, and 3-year annual spot rates from the Treasury prices.

Answer. The 1-year zero gives 4.00%. Using the 2-year coupon bond gives about 4.72%. Using the 3-year coupon bond after discounting earlier coupons gives about 5.14%.

1.23 Question 23 (Spreadsheet)

Open medium_spread_duration.csv. Estimate dollar loss for each corporate bond if spreads widen by 40 bps, using spread duration.

Answer. Approximate loss = -spread duration x market value x spread change. The losses are about $19,200, $30,000, and $14,400; total portfolio loss is about $63,600.

1.24 Question 24 (Spreadsheet)

Open medium_tbill_quotes.csv. Compute price and bond-equivalent yield for each bill.

Answer. Using the bank discount formula and then BEY formula, the approximate prices are $9,888.75, $9,793.33, and $9,498.33. The corresponding BEYs are about 4.56%, 4.25%, and 5.16%.

1.25 Question 25 (Spreadsheet)

Open medium_mbs_cashflows.csv. Compute SMM, prepayment, total principal, and ending balance for each month.

Answer. Apply SMM = 1 - (1 - CPR)^(1/12), prepayment = SMM x balance after scheduled principal, total principal = scheduled principal + prepayment, and ending balance = beginning balance - total principal. The ending balances are approximately $1,192,330, $1,184,071, and $1,174,985.